

Saksham Ojha

UG (IV Year I Semester)
B.Tech. (Metallurgical & Materials Engineering)
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Indian Institute of Technology Roorkee

VERIFIED

Area of Interest

Data Science, Machine Learning, Deep Learning, Gen AI, Product

Education

Year	Degree/Examination	Institution/Board	
2026	B.Tech. 3rd Year	Indian Institute of Technology, Roorkee	
2022	Intermediate (Class XII)	OPG World School (CBSE)	95.00 %
2020	Matriculate (Class X)	Mount Carmel School (CBSE)	97.40 %

Internships

AI/ML Engineering Intern | Edden Internet Pvt. Ltd. December 2024 - January 2025

- Contributed to **end-to-end automation** of short-form video creation using **GenAI** , optimizing research, scripting, and synthesis to streamline content production workflows.
- Built and integrated **ML pipelines** to personalize content recommendations, enhance audience relevance, and boost engagement across multiple digital platforms.
- Collaborated with **cross-functional teams** to streamline content workflows, reduce manual effort, and scale **AI-powered delivery** through efficient automation and process improvements.

Data Analytics Intern | SR Medicare Pvt. Ltd. May 2025 - June 2025

- Analysed **inventory, costing, and procurement data** to identify **trends** , evaluate **cost drivers** , and support **data-driven purchasing decisions** .
- Prepared **data analysis reports** and **automated MIS dashboards** to track **sales performance** , monitor **key business metrics** , and communicate **actionable insights** .
- Analysed **inventory levels and export timelines** to identify **supply chain bottlenecks** and support improvements in **inventory planning** and **operational efficiency** .

Growth Intern | Vary Gaming March 2025 - April 2025

- Boosted reach by **20%** through digital campaigns, partnering with cross-functional teams across marketing and product.
- Enhanced campaign performance and product strategy by analyzing **5+ KPIs** using Google Analytics and Excel.
- Refined brand positioning by identifying growth opportunities through market and competitor research across 3 verticals.

AI Engineering Intern | Jastro July 2026 - August 2026

- Developed AI solutions for personalized digital reports, leveraging LLMs to generate and refine user-specific content.
- Optimized AI workflows for automated content generation and evaluation: improving quality, consistency, and scalability.

Founder's Office Intern | SuperErgo May 2026 - June 2026

- Conducted **market research** and **competitor benchmarking** to identify growth opportunities, assess market positioning, and inform strategic decisions.
- Analyzed **business performance** and **operational bottlenecks** , translating findings into actionable recommendations to improve efficiency and support growth.

Product Intern | Jastro December 2025 - February 2026

- Analyzed **user journeys** across astrology products to identify **friction points** in onboarding, personalized report discovery, and content consumption; proposed **product improvements** to enhance user experience.
- Collaborated on **product requirements** and **feature prioritization** for personalized astrology reports, translating **user needs** into actionable product enhancements.
- Researched **competitor astrology platforms** , comparing **features, personalization, and monetization strategies** to identify opportunities for **product differentiation** and **user engagement** .

Projects

Docuquery- Semantic Query answering using RAG | Tinkering Lab IITR April 2025 - May 2025

- Built **OCR-based PDF text extraction** using binarization for improved accuracy & efficient chunking for embeddings.
- Implemented **semantic search** using all-mpnet-base-v2 embeddings and FAISS for fast document retrieval.
- Integrated a **T5-large transformer** to generate real-time, context-aware answers from retrieved PDF content.

3D Residual Stress Calculations | MMED IITR April 2025 - May 2025

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TMI Project: Introduction to Basic Water Quality Assessment | Hydrology Dept. IITR September 2023 - November 2023

- Performed water quality assessment of tap, canal, and sewage sources using **grab sampling** and advanced instruments.
- Measured **key parameters** like pH, DO, EC, turbidity to assess potability and recommend appropriate treatment actions.

Blockbase Project: On-Chain Expense Tracker Dapp | Finance & Economics Club, IIT Guwahati April 2025

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Sales Analysis using SQL and Power BI | Finance & Economics Club, IIT Guwahati August 2024

- Built interactive Power BI dashboards analyzing data from **100+** suppliers to uncover key business trends
- Developed **20+ KPIs** (cost/unit, margins, supplier performance) using SQL and Python
- Automated data workflows, reducing manual effort by **80%** and enabling real-time decision-making

Fundamental Analysis Case Study | Finance & Economics Club, IIT Guwahati August 2024

- Completed a **supply chain** project covering procurement, logistics, and inventory, with focus on ERP, AI and IoT tools.
- Focused on KPIs, cost optimization and **risk mitigation** to improve operational efficiency and customer fulfillment.

Open Project : Optimizing Air Travel Through Flight Delay Prediction | Society of Business, IIT Roorkee June 2025

- Analysed **600K+ flight records** to identify delay patterns and correlations between **weather conditions, flight operations, and departure punctuality** .
- Engineered predictive features and developed **machine learning models using Scikit-learn** to classify flight delays and identify key drivers of disruptions.
- Evaluated **model performance** and visualised predictive insights to support **delay-risk assessment, operational planning, and flight disruption mitigation** .

Credit Card Score Modelling via Classification & Risk Techniques | Finance Club, IIT Roorkee May 2025 - June 2025

- Built a credit risk model using **Logistic Regression** and **Random Forest** balancing a **22% default class** with SMOTE.
- Achieved **81% recall** , **0.615 F2-score** , and **0.789 AUC-ROC** and reduced false negatives(FN) by **40%** .

Remind - Prescription-Based Medicine Reminder | Tinkering Lab IITR January 2025 - April 2025

- Built a Flutter app to **auto-schedule** medicine reminders from prescriptions using notifications and background tasks.
- Designed a secure, intuitive UI with pill tracking, refill alerts and dosage compliance features for improved adherence.

LLM based Health Assistant | Tinkering Lab IITR May 2025 - June 2025

- Built a **RAG** pipeline using **LangChain, Pinecone** and **GPT-3.5** for context-aware Q&A over indexed medical literature.
- Applied search using dense embeddings and async pipelines improving retrieval and response quality in realtime.

Automated Meta Data Generation | MaRS IITR June 2025

- Automated metadata generation for unstructured documents using **NLP techniques** (**spaCy**, **Transformers**, topic modeling, document embeddings) to extract key entities, keywords, and topics.
- Deployed scalable Python pipelines for **semantic indexing**, improving content discoverability, metadata accuracy, and search relevance across diverse datasets.

Refined Knowledge Distillation | Tinkering Lab IITR January 2025

- Implemented a **teacher-free** distillation model using self-training and label smoothing on CIFAR-100 and ImageNet.
- Improved **CNN model generalization** (ResNet, ResNeXt, MobileNet, ShuffleNet) without external teacher models.
- Achieved up to **+1% accuracy gain** using optimized training pipelines and **PyTorch** based deep learning workflows.

Finetuning Deep Learning Models on Astrophysical Datasets | PaAC, IIT Roorkee September 2024 - November 2024

- Achieved **79.6% accuracy** using **ResNet/VGG** with transfer learning; experimented with **LoRA fine-tuning.**
- Reduced model loss by **4.7%** and improved interpretability using **Grad-CAM** and evaluated via confusion matrix.

Meesho: Business & Growth Strategy Analysis | Society of Business, IIT Roorkee March 2026 - April 2026

- Analysed **Meeshos zero-commission business model** , examining how fulfilment, advertising, and content commerce generated **12,626 crore in FY26 revenue** .
- Evaluated **growth across 264M users and 2.67B orders** , benchmarking Meeshos value-led strategy against **Amazon, Flipkart, and quick-commerce platforms** .
- Assessed **FY26 investments in Valmo and 990 crore in new-user acquisition** , analysing their implications for **operating losses and cohort-level profitability** .

Stable Money: FinTech GTM Strategy & Market Entry Analysis | Society of Business, IIT Roorkee April 2026 - May 2026

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Intelligent SQL Query Agent | Hostel Council, IIT Roorkee May 2026 - June 2026

- Built an in-browser SQL agent that locks metric, join path, and data grain before query execution.
- Implemented a query critic to execute and compare naive versus defended SQL against SQLite/WASM.
- Developed a Next.js/TypeScript interface with deterministic routing, sensitivity analysis, and auditable row-level results.

Impact of Global Conflicts on Crude Oil Market | Finance Club IIT Roorkee May 2026 - June 2026

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Digital Finance Adoption in Rural India | NSS IIT Roorkee June 2026

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Support Integrity Auditor | MaRS IIT Roorkee June 2026

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Direct Browser-to-Browser File Transfer Platform | MaRS IIT Roorkee June 2026

- Built encrypted P2P file transfer using chunking, encryption, and reassembly over WebRTC data channels.
- Implemented Socket.IO + Express signaling with STUN/ICE NAT traversal and real-time connection tracking.

Agent-Based Dynamic Tariff Optimization for EV Charging Networks | Society of Business, IIT Roorkee May 2026 - June 2026

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Redefining the Mobile Experience Through Generative AI | Society of Business, IIT Roorkee May 2026 - June 2026

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Verifiable-RL: GRPO for Long-Horizon Tool Agents with Reward-Hack Controls | Vision & Language Group, IITR June 2026 - July 2026

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Echo: Real-Time Voice Agent Platform | Vision & Language Group, IITR February 2026 - March 2026

- Built a duplex voice agent with WebRTC, speech recognition, tool routing, and speech synthesis.
- Implemented barge-in cancellation across agent generation, TTS, tools, and Twilio audio streams.
- Developed a Hinglish evaluation bench measuring tool accuracy, time-to-first-audio, and barge-in latency.

EffectEval: Agent Evaluation & Tool-Use Benchmark | Vision & Language Group, IIT Roorkee June 2026 - July 2026

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Confera | Rajiv Bhawan, IITR May 2026 - June 2026

- Built a multi-agent event planner using FastAPI, WebSockets and ChromaDB for workstreams, state and retrieval.
- Developed a React + TypeScript dashboard with Vite and Zustand for real-time progress, review, and event planning.

Local LLM Instrumentation, Tracing and Replay Platform | GDSC IITR June 2026 - July 2026

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Together: Group Coordination Made Simple | Self Project July 2026 - August 2026

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Awards / Scholarships / Academic Achievements

- Qualified **NTSE** - Secured rank within **top 2,000** out of **1 million+** candidates nationwide.
- Awarded **Certificate of Merit** - Placed within **top 300** in Indian Olympiad Qualifier in Mathematics **(IOQM)-2020.**
- **Passed with Merit** in Preliminary English Test (Council of Europe Level B1) conducted by **University of Cambridge.**
- Secured **99.68 percentile** in JEE Mains 2023 - Ranked **3757** out of **1.2Million** candidates nationwide.
- Qualified **JSTSE** - Secured **Rank 24** among **50,000+** candidates in Junior Science Talent Search Exam
- **2nd Runner-Up** : Gen AI Exchange Hackathon 2025 by **Google Cloud**
- **Winner** The IndeHub Hackathon 2025
- Codeforces Rating: 1187

Skills

Computer languages	Python, C++ , SQL
Software Packages	NumPy, Pandas, scikit-learn, Matplotlib, Streamlit, Scipy, PyTorch, Transformers, Diffusers, LangChain, XGBoost, LightGBM, CatBoost, Computer Vision
Additional Courses	Supervised Machine Learning by Andrew Ng, Neural Networks by Andrew Ng
Languages Known	English, Hindi

Positions of Responsibility & Extra Curriculars

Executive Member | Tinkering Lab IITR May 2025 - Present

- Executed multiple software, product and AI/ML projects, both independently and in **collaborative** team environments.
- Scored full marks in the **OpenCV-based** assignment in **e-Yantra Robotics Competition** organized by **IIT Bombay.**
- Coordinated national-level hackathons including **Tinkerquest** , engaging **1,000+** participants from across India.
- Judged and coordinated Crawl-o-tron and Nano Navigator at Cognizance, **Asias second-largest** student-run tech fest.

Member | Watch Out IITR December 2024 - Present

- Spearheaded **editorial planning** and publication of reports covering academics, placements, and campus affairs.
- Enhanced campus engagement and transparency through collaboration, **impactful journalism** and strategic content.

Event Coordinator | Cognizance March 2025

- Judged and coordinated Crawl-o-tron and Nano Navigator at **Cognizance 2025** - **Asias second-largest** student-run technical fest with **3,000+ participants** from across India and abroad.

Volunteer | Thomso October 2024

- Volunteered for the seamless execution of the AR Treasure Hunt at **Thomso24** , one of Indias largest cultural festivals, with **10,000+** participants from over **200+ colleges** nationwide.

Participant | e-Yantra, IIT Bombay August 2024 - December 2024

- Developed an **Automated Arena Mapping** and Obstacle Detection using **ArUco Markers** and Image Processing

Maintenance Secretary | Rajiv Bhawan, IITR November 2025 - Present

- Represented **650+ residents** , bridging communication with administration and resolving key student issues.
- Spearheaded infrastructure upgrades, managing a **Rs. 1M+** welfare fund to enhance living for residents.
- Optimized a **Rs. 250K+** budget via competitive negotiations to procure sports inventory and custom jerseys.

Web Developer | Student Technical Council, IITR December 2025 - Present

- Developed **Srishti** hackathon website, optimizing performance through code splitting, lazy loading, and memoization.
- Coordinated technical events and supported 20+ student clubs, managing their technical activities and digital initiatives.

References

Dr. Kamal
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